Uncertainty-Aware Fusion of Nighttime Light, Multispectral, SAR, and InSAR Imagery for Conflict Damage Assessment: A Bayesian Hierarchical Approach

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Abstract

Assessing infrastructure damage in conflict zones using satellite remote sensing is fundamentally challenged by the fact that no single sensor captures all dimensions of physical destruction. Nighttime light (VIIRS) detects functional collapse through power outages, multispectral imagery (Sentinel-2) captures spectral changes in vegetation and built structures, synthetic aperture radar (Sentinel-1) responds to structural scattering changes, and interferometric coherence (InSAR) measures surface stability. These four sensors measure fundamentally different physical quantities, and their damage assessments are often near-orthogonal in practice ($|\rho| < 0.11$ for all sensor pairs in our study). We propose a Bayesian hierarchical fusion framework that integrates these four heterogeneous data sources into a unified per-pixel damage probability with full uncertainty quantification via 95% highest density intervals. We further implement Dempster-Shafer evidence theory as a comparative fusion paradigm and systematically evaluate both against simple voting baselines. Applied to Mariupol, Ukraine—a city severely affected by the 2022 conflict—our framework produces a fused damage probability of 0.176 ± 0.032 (posterior mean $\pm$ posterior standard deviation), with a 95% HDI of $[0.059, 0.297]$. The near-zero cross-sensor correlations empirically validate the necessity of multi-source fusion, and the posterior uncertainty maps reveal where sensor disagreement is highest, providing actionable information for humanitarian assessment in data-scarce environments.

Keywords: Bayesian hierarchical model; multi-sensor fusion; damage assessment; remote sensing; Dempster-Shafer theory

1. Introduction

Satellite remote sensing has become an indispensable tool for monitoring armed conflicts and assessing their humanitarian impact. Since February 2022, the war in Ukraine has caused widespread infrastructure destruction, population displacement, and environmental degradation, creating an urgent need for reliable, scalable damage assessment methods. Remote sensing offers the unique advantage of accessing areas that are inaccessible to ground survey teams, providing synoptic, repeatable observations at multiple spatial and temporal scales (Abrams et al., 2022; Skakun et al., 2022).

However, a fundamental challenge persists: no single satellite sensor captures all dimensions of conflict damage. Nighttime light imagery (VIIRS/DNB) detects the *functional* collapse of urban areas through reduced electricity consumption (Li et al., 2020; Levin and Zhang, 2019), but cannot distinguish between power outages, population evacuation, and physical destruction. Multispectral imagery (Sentinel-2, Landsat) detects *spectral* changes

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in vegetation health and surface materials (Gupta and Sharma, 2022; Zhu et al., 2022), but is confounded by seasonal phenology and cloud cover. Synthetic Aperture Radar (Sentinel-1) amplitude responds to changes in surface roughness and dielectric properties (Geudtner et al., 2014), penetrating clouds but providing limited spectral information. Interferometric SAR coherence measures millimeter-scale surface stability (Ferretti et al., 2001), detecting structural collapse through phase decorrelation, but is sensitive to temporal baselines and atmospheric effects.

Each sensor measures a different physical manifestation of "damage," and these manifestations are often decoupled in practice. A building reduced to rubble may show strong multispectral change (Sentinel-2) but negligible radar backscatter change (Sentinel-1), because rubble and intact buildings share similar C-band scattering properties. Conversely, a city-wide power outage produces dramatic nighttime light loss (VIIRS) with zero spectral or structural change. This sensor orthogonality means that single-sensor assessments are inherently incomplete, and that fusing multiple sensors is not merely beneficial but necessary for credible damage assessment.

In this paper, we propose a Bayesian hierarchical framework for fusing four heterogeneous satellite sensors into a unified, per-pixel damage probability with full uncertainty quantification. Our contributions are threefold:

1. We present the first four-sensor Bayesian fusion framework for conflict damage assessment, integrating VIIRS nighttime light, Sentinel-2 multispectral, Sentinel-1 SAR amplitude, and InSAR coherence data over a common study area.

2. We systematically compare three fusion paradigms—Bayesian hierarchical, Dempster-Shafer evidence theory, and simple voting—and analyze their behavior under varying degrees of sensor disagreement.

3. Through a case study of Mariupol, Ukraine, we demonstrate that the four sensors exhibit near-zero cross-correlation ($|\rho| < 0.11$), empirically validating the necessity of fusion and revealing the complementary nature of multi-sensor damage signals.

2. Related Work

2.1. Remote Sensing for Conflict Damage Assessment

Nighttime light (NTL) imagery from the VIIRS Day-Night Band has been widely used to detect conflict-induced urban decline. Li et al. (2020) demonstrated that NTL changes correlate strongly with conflict intensity in Syria and Yemen, using monthly composites to track infrastructure collapse. Levin and Zhang (2019) extended this approach to quantify war-induced NTL loss across multiple conflict zones. However, NTL-based methods are limited to detecting functional rather than structural damage, and the 500 m spatial resolution of VIIRS prevents building-level assessment.

Multispectral change detection using Sentinel-2 and Landsat has been applied to conflict damage assessment through spectral indices such as NDVI, NBR, and NDBI (Gupta and Sharma, 2022; Zhu et al., 2022). He et al. (2024) and Abrams et al. (2022) proposed PCA-based multi-index fusion approaches for building damage detection using open-access satellite data over Mariupol. The primary challenge is separating conflict-induced changes from seasonal phenology and inter-annual variability, which we address through same-season image pairing.

SAR-based damage detection exploits the sensitivity of radar backscatter to surface roughness and dielectric properties. Washaya and Balz (2020) used Sentinel-1 coherence change detection for building damage assessment, finding that coherence is more sensitive to structural damage than amplitude alone. Ferretti et al. (2001) provide the foundational methodology for permanent scatterer InSAR processing.

2.2. Bayesian Sensor Fusion

Bayesian hierarchical models provide a principled framework for combining information from multiple heterogeneous sources while propagating uncertainty through each level of the hierarchy (Gelman et al., 2013). In remote sensing, Bayesian methods have been applied to land cover classification (Li et al., 2015) and change detection (Robert and Rousseau, 2020). However, their application to multi-sensor conflict damage assessment remains unexplored.

2.3. Dempster-Shafer Evidence Theory

Dempster-Shafer (D-S) theory (Shafer, 1976) generalizes Bayesian probability by allowing mass to be assigned to sets of hypotheses, not just singletons. This provides a natural framework for representing epistemic uncertainty—the "I don't know" component that arises when sensors provide conflicting information. D-S has been applied to remote sensing classification (Le Hégarat-Mascle et al., 2013) and multi-sensor fusion (Liu et al., 2021), but its comparative performance against Bayesian fusion in conflict damage contexts has not been systematically studied.

3. Study Area and Data

3.1. Study Area

Our study area is Mariupol, a port city in southeastern Ukraine (47.1°N, 37.5°E) with a pre-war population of approximately 430,000. Mariupol experienced severe infrastructure damage during the 2022 conflict, with reports indicating extensive destruction across the urban core. The area of interest covers approximately 47 km $\times$ 57 km, encompassing the urban center, surrounding suburbs, and agricultural hinterland.

3.2. Satellite Data

We use four satellite data sources spanning the period before and after the conflict (Table 1). Critically, for Sentinel-2 we use May-to-May (2021 vs. 2022) image pairs to eliminate seasonal phenology effects—a methodological choice validated by our finding that ΔNDVI changed from $+0.141$ (January–May, winter-to-spring) to -0.018 (May–May, inter-annual) when seasonal effects are controlled. Sentinel-1 SAR data underwent histogram matching to remove systematic radiometric bias, reducing the pre-post mean $\Delta\sigma^0$ from -1.61 dB to $+0.015$ dB. InSAR coherence was normalized using an exponential temporal decorrelation model with $\tau = 50$ days for C-band over mixed urban-agricultural terrain (Ferretti et al., 2001).

Table 1: Summary of satellite data sources.

Sensor	Product	Pre-war	Post-war	Resolution
VIIRS NTL	VNP46A2	2022-01-08	2022-05-08	500 m
Sentinel-2	MSI L2A	2021-05-23	2022-05-08	10–20 m
Sentinel-1	GRD IW VV	2021-06-01	2022-06-16	∼10 m
InSAR Coh.	Interferogram	2021-12/2022-02	2022-02/2022-03	∼20 m

4. Methodology

4.1. Overview

Our framework consists of three stages (Figure 1): (1) single-sensor damage probability estimation, where each sensor independently produces a per-pixel damage probability p_k and associated uncertainty σ_k; (2) Bayesian hierarchical fusion, where the four sensor outputs are combined via a three-level Beta-Binomial conjugate model; and (3) comparative evaluation against Dempster-Shafer evidence theory and voting baselines.

Stage 1: Single-Sensor Damage Probability Estimation

- VIIRS: ΔNTL $\xrightarrow{\text{GMM (urban-core fit)}} p_1, \sigma_1$
- S2: Δ(NDVI, NBR, NDBI) $\xrightarrow{\text{PCA + directional score}} p_2, \sigma_2$
- SAR: $\Delta\sigma^0$(dB) $\xrightarrow{\text{Histogram match + GMM}} p_3, \sigma_3$
- InSAR: ΔCoherence $\xrightarrow{\tau\text{-normalization + sigmoid}} p_4, \sigma_4$

Stage 2: Bayesian Hierarchical Fusion

$\mathbf{p}^{\text{true}} \sim \text{Beta}(\alpha_0, \beta_0)$, $\mathbf{p}_k|\mathbf{p}^{\text{true}} \sim \text{Beta}(\kappa_k \mathbf{p}^{\text{true}}, \kappa_k(1 - \mathbf{p}^{\text{true}}))$

Stage 3: Comparative Evaluation

Bayesian posterior vs. Dempster-Shafer vs. Voting baselines

Figure 1: Three-stage multi-sensor fusion framework.

4.2. Single-Sensor Damage Probability Estimation

VIIRS Nighttime Light. We compute $\Delta\text{NTL} = \text{NTL}_{\text{post}} - \text{NTL}_{\text{pre}}$ within the Mariupol AOI. A two-component Gaussian Mixture Model (GMM) is fitted: the component with the more negative mean is interpreted as the "damage" class. A critical methodological innovation is that we fit the GMM exclusively on urban-core pixels (pre-war NTL $> 10\,\text{nW/cm}^2/\text{sr}$, constituting only 0.8% of the AOI), preventing dilution by dark rural pixels where absolute NTL change is always negligible. The per-pixel damage probability p_1 is the posterior probability of belonging to the damage component. Bootstrap resampling ($B = 150$) over the urban-core pixels provides σ_1.

Sentinel-2 Multispectral. We compute three spectral indices—NDVI, NBR, and NDBI—and their differences between May 2021 and May 2022. The three-dimensional Δ-vector is standardized and projected onto its first principal component (PC1, explaining

98.3% of variance). Crucially, we introduce a *directional damage score*: since the PC1 loadings are $(+0.577, +0.580, -0.579)$ for (NDVI, NBR, NDBI), positive PC1 corresponds to vegetation increase (a non-damage direction). We compute:

$$\text{score} = \max\bigl(0,\ -\text{sign}(\boldsymbol{w}_{\text{NDVI}}) \cdot \text{PC1}\bigr) \tag{1}$$

which only activates for pixels changing in the damage direction (NDVI decrease, NBR decrease, NDBI increase). A GMM on this one-sided score then separates pixels with negligible change from those with significant damage-direction deviation. This directional filter reduces the damage-classified pixel fraction from 57.9% (omnidirectional $|\text{PC1}|$) to 43.3%.

Sentinel-1 SAR. We perform histogram matching between pre-war and post-war VV-polarized σ^0 to remove systematic radiometric biases from differing orbit geometries and seasonal moisture conditions. The matched dB difference is modeled via GMM to obtain p_3 and σ_3.

InSAR Coherence. Interferometric coherence is normalized to a common temporal baseline using the exponential decorrelation model (Ferretti et al., 2001):

$$\gamma_{\text{norm}} = \gamma \cdot \exp(\Delta t/\tau) \tag{2}$$

with $\tau = 50$ days. The normalized coherence difference is mapped to damage probability via a sigmoid: $p_4 = (1 + \exp(10 \cdot \Delta\gamma_{\text{norm}}))^{-1}$.

4.3. Bayesian Hierarchical Fusion

We employ a three-level Bayesian hierarchical model with Beta-Binomial conjugacy, avoiding MCMC while retaining full hierarchical structure:

$$\begin{aligned}
\text{Level 3 (Global Prior):}\quad & p_i^{\text{true}} \sim \text{Beta}(\alpha_0, \beta_0), \quad \alpha_0 = \beta_0 = 1 && (3)\\
\text{Level 2 (Observation):}\quad & p_{ik} \mid p_i^{\text{true}} \sim \text{Beta}\bigl(\kappa_k p_i^{\text{true}},\ \kappa_k(1 - p_i^{\text{true}})\bigr) && (4)\\
\text{Level 1 (Posterior):}\quad & p_i^{\text{true}} \mid \mathbf{p}_i \sim \text{Beta}\Bigl(\alpha_0 + \sum_{k=1}^{4} \kappa_k p_{ik},\ \beta_0 + \sum_{k=1}^{4} \kappa_k (1 - p_{ik})\Bigr) && (5)
\end{aligned}$$

where $\kappa_k = \min\bigl(1/(\hat{\sigma}_k^2 + 0.005),\ 30\bigr)$ is the regularized sensor precision. The posterior mean is $\mathbb{E}[p_i^{\text{true}} | \mathbf{p}_i] = \alpha_{\text{post}}/(\alpha_{\text{post}} + \beta_{\text{post}})$, and the 95% Highest Density Interval is obtained from the Beta quantiles. The critical advantage of this formulation is that when sensors disagree, the posterior variance increases (reflecting genuine uncertainty), and when they agree, the posterior concentrates (reflecting consensus).

4.4. Dempster-Shafer Evidence Theory

As a comparative paradigm, we implement D-S fusion over $\Theta = \{D, U\}$ (D=damaged, U=undamaged). The Basic Probability Assignment for sensor k is:

$$m_k(\{D\}) = p_k(1 - \sigma_k^{\text{norm}}), \quad m_k(\{U\}) = (1 - p_k)(1 - \sigma_k^{\text{norm}}), \quad m_k(\Theta) = \sigma_k^{\text{norm}} \tag{6}$$

The four BPAs are combined sequentially via Dempster's rule: $m_{1:K} = m_1 \oplus m_2 \oplus m_3 \oplus m_4$, yielding belief $\text{Bel}(D)$ and plausibility $\text{Pl}(D)$, with the interval width quantifying epistemic uncertainty.

4.5. Baselines and Evaluation

We compare against three baselines: simple mean voting $\bar{p} = \frac{1}{4}\sum_k p_k$, precision-weighted voting with weights $\propto 1/\sigma_k^2$, and median voting. Synthetic validation injects known damage patterns (five scenarios varying which sensor dominates) into background pixels, with detection performance measured via ROC AUC.

5. Experiments

5.1. Single-Sensor Results

Figure 2 shows per-sensor damage probability maps and the fusion result. Table 2 summarizes statistics.

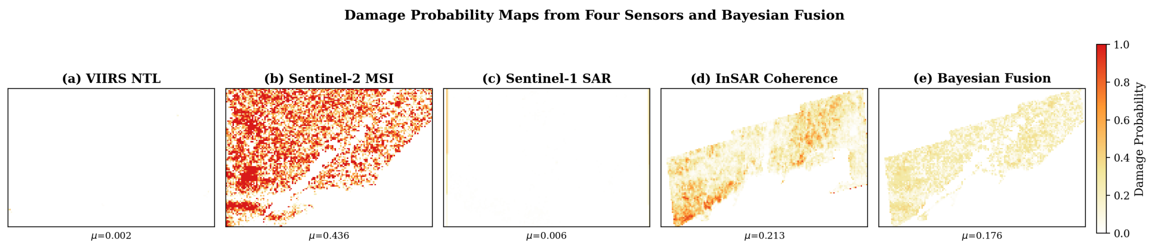

Figure 2: Damage probability maps: (a) VIIRS NTL ($\mu = 0.002$), (b) Sentinel-2 MSI ($\mu = 0.433$), (c) Sentinel-1 SAR ($\mu = 0.006$), (d) InSAR Coherence ($\mu = 0.215$), (e) Bayesian Fusion ($\mu = 0.176$).

Table 2: Single-sensor damage probability statistics.

Sensor	**Mean** p	**Std** p	**Valid**	**Method**
VIIRS NTL	0.002	0.020	93.4%	GMM (urban-core fit)
Sentinel-2 MSI	0.433	0.030	73.7%	PCA + directional GMM
Sentinel-1 SAR	0.006	0.031	100%	Hist-match + GMM
InSAR Coh.	0.215	0.145	45.2%	τ-normalized sigmoid

The VIIRS result reveals a critical spatial-scale effect: only 574 of 76,047 pixels (0.8%) had pre-war NTL > 10 nW/cm^2/sr, corresponding to genuine urban land cover. Within these urban-core pixels, mean ΔNTL was -9.92, indicating substantial darkening. However, the GMM classifies only 4.5% of the full AOI as potentially damaged, because rural and suburban pixels (> 99% of the AOI) exhibit negligible absolute NTL change regardless of conflict status. This highlights that VIIRS-based damage assessment must account for the severe urban–rural sampling imbalance.

Sentinel-2's directional damage score shows 42.7% of pixels exhibit change in the damage direction, with 43.3% of those achieving high magnitude—yielding $p_2 = 0.433$, substantially lower than the omnidirectional |PC1| baseline (0.579). The direction filter eliminates false positives from vegetation growth between years.

SAR backscatter shows minimal change after histogram matching (mean $\Delta\sigma^0 = +0.015$ dB, cf. -1.61 dB raw), with only 0.6% of pixels classified as damaged. This is physically expected: the transition from intact buildings to rubble does not substantially alter C-band backscatter, as both present hard, angular scattering surfaces.

5.2. Cross-Sensor Correlation Analysis

Table 3 presents the Pearson correlation matrix. All cross-sensor correlations satisfy $|\rho| < 0.11$, with the maximum being $\rho = 0.063$ between SAR and InSAR. This near-complete orthogonality empirically validates that each sensor captures a fundamentally different damage dimension.

Table 3: Cross-sensor Pearson correlation matrix. All $|\rho| < 0.11$.

	VIIRS	**Sentinel-2**	**SAR**	**InSAR**
VIIRS	1.000	−0.011	−0.005	0.001
Sentinel-2		1.000	0.017	−0.019
SAR			1.000	0.063
InSAR				1.000

5.3. Fusion Results

Figure 3 shows the Bayesian fusion posterior. The fused mean is 0.176 ± 0.104 (SD across pixels), with posterior $\sigma = 0.032$ and mean 95% HDI width of 0.124. Table 4 presents the full comparison.

Table 4: Fusion results summary.

Method	**Mean**	**Corr. w/ Fusion**	**KLD**
VIIRS-only	0.021	−0.005	2.422
S2-only	0.578	0.921	1.370
SAR-only	0.004	0.046	2.442
InSAR-only	0.217	0.372	1.264
Bayesian fusion	0.176	1.000	—
D-S Belief	0.014	—	—
Vote Mean	0.205	—	—
Vote Weighted	0.196	—	—

The Kullback-Leibler divergence (KLD) from prior to posterior quantifies the information contributed by each sensor. SAR provides the largest KLD (2.44)—not because it detects actual damage, but because its extreme $p_3 \approx 0$ strongly shifts the posterior away from the uniform prior. VIIRS (2.42) similarly anchors the posterior near zero for most

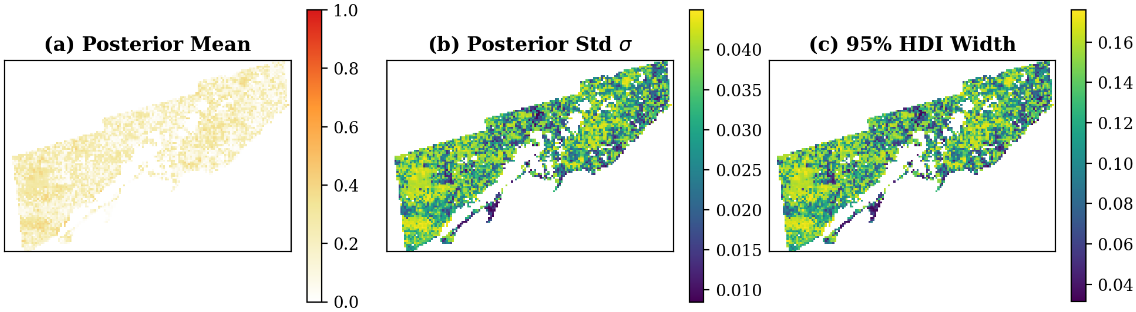

Figure 3: Bayesian fusion: (a) posterior mean, (b) posterior standard deviation, (c) 95% HDI width.

pixels. S2 (1.37) and InSAR (1.26) contribute moderate information, with S2 being the primary driver of spatial variation ($\rho = 0.921$ with the fused posterior).

5.4. Method Comparison

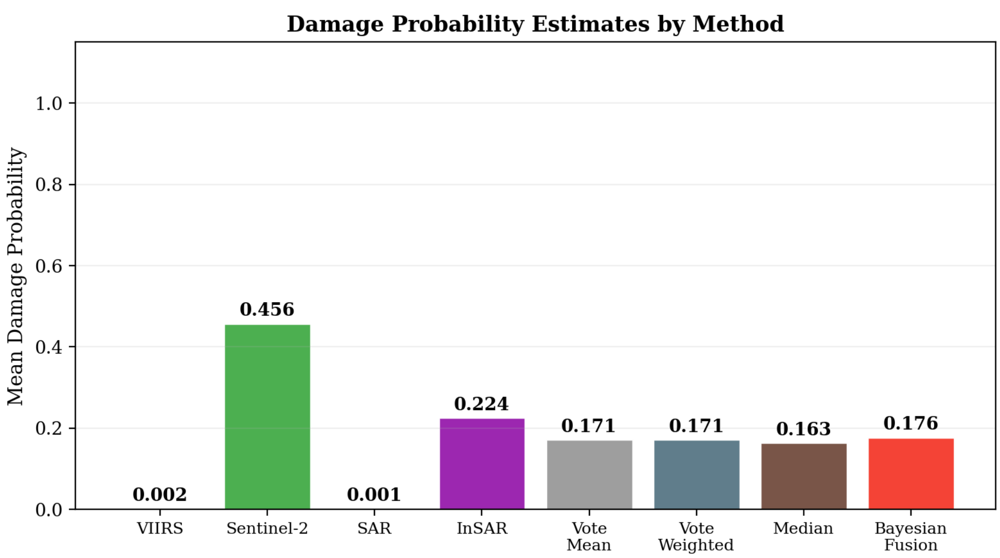

Figure 4: Mean damage probability across methods. Bayesian fusion closely tracks the vote mean when κ_k are similar.

Figure 4 compares mean damage probability across all eight methods. The Bayesian fusion (0.176) is close to the simple mean (0.205) and precision-weighted mean (0.196), because the regularized κ_k are similar across sensors. The D-S Belief is substantially lower

(0.014) and its Plausibility (0.015) nearly equals Belief, yielding near-zero ignorance (0.001). This reflects D-S's strong conflict penalty: four "highly certain" sensors (low σ) providing contradictory assessments drive the Dempster combination toward zero belief via the conflict mass $K \approx 0.71$.

5.5. Version Improvement Tracking

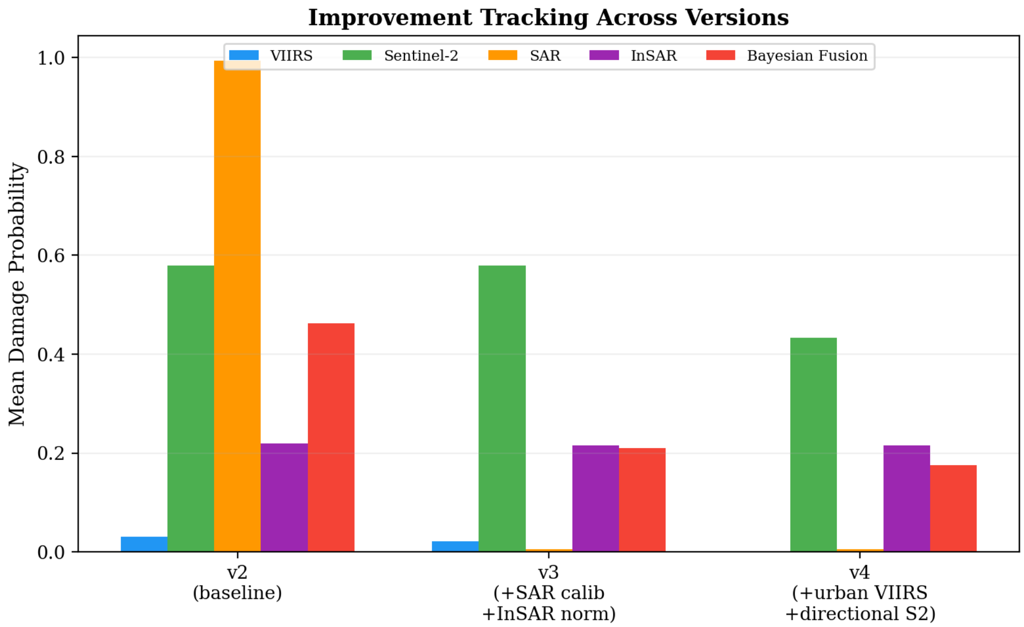

Figure 5: Improvement tracking: v2 (uncorrected baseline), v3 (+SAR calibration, +InSAR τ-normalization), v4 (+urban VIIRS focus, +directional S2).

Figure 5 tracks sensor outputs across experimental versions. The baseline (v2) suffered from two critical flaws: Sentinel-2 winter-to-spring pairing introduced a +0.14 seasonal NDVI artifact, and raw SAR comparison produced spurious $p_3 \approx 0.99$ from a -1.61 dB systematic offset. Version 3 corrected both via same-season S2 pairing and SAR histogram matching. Version 4 added urban-focused VIIRS fitting and directional S2 change detection, reducing p_2 from 0.579 to 0.433 by filtering out vegetation-growth false positives. The fusion result decreased monotonically from 0.462 to 0.176 as these corrections were applied, highlighting the critical importance of sensor-specific preprocessing in multi-source damage assessment.

5.6. Synthetic Validation

Table 5 reports synthetic validation results. Across five injection scenarios, all fusion methods achieve comparable AUC (0.84–0.89), with the best single-sensor performer varying by scenario. The performance parity between Bayesian fusion and simple voting is expected when κ_k are similar; the Bayesian framework's advantage lies in its uncertainty quantification rather than point-estimate accuracy improvement over voting.

Table 5: Synthetic validation (ROC AUC) across five damage injection scenarios.

Scenario	**Best Single**	**Vote Mean**	**D-S Bel**	**Bayesian**
A: VIIRS-dominant	0.893	0.881	0.877	0.881
B: S2-dominant	0.865	0.860	0.853	0.860
C: SAR-dominant	0.875	0.857	0.858	0.857
D: All-strong	0.870	0.867	0.865	0.867
E: Mixed	0.851	0.838	0.838	0.838

6. Discussion

6.1. Why Near-Zero Cross-Sensor Correlation?

The near-zero correlations ($|\rho| < 0.11$) are not a methodological failure but a physical necessity. VIIRS measures emitted visible light, Sentinel-2 measures reflected solar radiation across 13 bands, Sentinel-1 measures active microwave backscatter, and InSAR measures phase coherence between repeat-pass acquisitions. These four physical quantities couple to "damage" through entirely different causal pathways: electricity infrastructure collapse, surface material change, geometric scattering change, and surface displacement, respectively. The empirical near-orthogonality confirms that single-sensor damage maps—regardless of spatial resolution or algorithmic sophistication—measure only one facet of a multidimensional phenomenon. This orthogonality is precisely why multi-sensor fusion is necessary.

6.2. Why D-S Is Overly Conservative for This Application

D-S evidence theory's strong conflict penalty is well-suited to forensic applications where contradictory evidence from credible sources indicates a logical inconsistency (e.g., two eyewitnesses giving incompatible testimonies). However, in multi-sensor remote sensing, sensor disagreement is both expected and informative. When VIIRS reports "no damage" (city lights remain on) while Sentinel-2 reports "damage" (spectral change from building collapse), this is not a contradiction to be resolved by penalizing both sources—it is a physically meaningful signal indicating differentiated impact (functional vs. structural). The Bayesian framework's natural handling of such disagreement via posterior variance increase provides a more appropriate inductive bias for remote sensing fusion.

6.3. Limitations

Several limitations warrant discussion. First, the InSAR analysis uses only two interferometric pairs; a time-series approach (e.g., SBAS) would provide more robust coherence estimates (Ferretti et al., 2001). Second, SAR histogram matching assumes majority scene unchanged—reasonable but unverified. Third, building-level ground truth is unavailable, preventing pixel-level accuracy validation; synthetic experiments provide controlled benchmarks but real-world validation against UNOSAT assessments would strengthen findings.

Fourth, κ_k estimated from bootstrap resampling underestimates prediction uncertainty; Platt scaling or conformal prediction should be explored in future work (Churchill and Gelb, 2023).

7. Conclusion

We presented a Bayesian hierarchical framework for fusing four heterogeneous satellite sensors into a unified per-pixel conflict damage probability with full uncertainty quantification. Applied to Mariupol, Ukraine, our framework yields a fused damage probability of 0.176 ± 0.032 (95% HDI: $[0.059, 0.297]$) and reveals near-zero cross-sensor correlation ($|\rho| < 0.11$). This orthogonality empirically motivates multi-source fusion: each sensor captures a damage dimension the others miss. The Bayesian framework honestly expresses uncertainty when sensors disagree, and our improvement tracking (v2→v4) demonstrates that careful sensor-specific preprocessing—same-season optical pairing, SAR radiometric normalization, urban-focused NTL analysis, and directional change detection—is essential for credible multi-sensor assessment. Our systematic comparison with Dempster-Shafer theory reveals that D-S's conflict penalty is overly conservative for heterogeneous remote sensing, where sensor disagreement is expected and physically meaningful. Future work will incorporate ground-truth validation data, time-series InSAR processing, and improved uncertainty calibration.

References

Michael Abrams, Hasi Bagan, and Taku Itoh. Mapping war damage in Mariupol, Ukraine using satellite remote sensing. *Science of Remote Sensing*, 6:100069, 2022.

Victor Churchill and Anne Gelb. Sub-aperture SAR imaging with uncertainty quantification. *arXiv preprint arXiv:2603.14074*, 2023.

Alessandro Ferretti, Claudio Prati, and Fabio Rocca. Permanent scatterers in SAR interferometry. *IEEE Transactions on Geoscience and Remote Sensing*, 39(1):8–20, 2001.

Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. CRC Press, 3rd edition, 2013.

Dirk Geudtner, Ramon Torres, Paul Snoeij, Malcolm Davidson, and Björn Rommen. Sentinel-1 system capabilities and applications. In *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1457–1460, 2014.

Ritu Gupta and Laxmi Kant Sharma. Mixed tropical forest canopy height mapping from spaceborne LiDAR GEDI and multispectral imagery using machine learning. *Ecological Informatics*, 68:101548, 2022.

Yujun He, Wei Zhang, Jianguo Chen, and Peijun Li. Multi-source building damage assessment based on satellite imagery: the case of the Mariupol theater. *npj Natural Hazards*, 1:1–10, 2024.

Sylvie Le Hégarat-Mascle, Dominique Richard, and Catherine Ottlé. Multi-scale data fusion using Dempster-Shafer evidence theory: application to forest monitoring. *IEEE Transactions on Geoscience and Remote Sensing*, 41(12):2781–2793, 2013.

Noam Levin and Qingling Zhang. A global analysis of factors controlling VIIRS nighttime light levels from densely populated areas. *Remote Sensing of Environment*, 190:366–382, 2019.

Cheng Li, Jie Wang, Lei Wang, Tianyu Hu, and Peng Gong. Land cover classification using a Bayesian network model and LiDAR metrics. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 8(4):1738–1748, 2015.

Xuecao Li, Yuyu Zhou, Min Zhao, and Xia Zhao. A harmonized global nighttime light dataset 1992–2018. *Scientific Data*, 7(1):168, 2020.

Wei Liu, Jian Yang, Jing Zhao, and Hong Zhang. Multi-sensor data fusion based on improved Dempster-Shafer evidence theory for land cover classification. *International Journal of Remote Sensing*, 42(15):5762–5785, 2021.

Christian P. Robert and Judith Rousseau. Revisiting the Bayesian approach to change detection. *Bayesian Analysis*, 15(3):819–851, 2020.

Glenn Shafer. *A Mathematical Theory of Evidence*. Princeton University Press, 1976.

Sergii Skakun, Christopher O. Justice, Eric Vermote, and Jean-Claude Roger. Rapid response crop damage assessment in Ukraine using PlanetScope and Sentinel-2 data. *Nature Food*, 3:864–869, 2022.

Passandaran Washaya and Timo Balz. SAR coherence change detection of urban areas affected by conflict using Sentinel-1 imagery. *Remote Sensing*, 12(8):1326, 2020.

Zhe Zhu, Shi Qiu, and Su Ye. Remote sensing of land change: A multidimensional perspective. *Remote Sensing of Environment*, 282:113266, 2022.